

Lesson 1: Setup & Your First App

FlutterLearn | Module 1: Flutter Foundations

What is Flutter?

Flutter is Google's open-source toolkit for building apps from a single codebase. Write your code once in the Dart language, and Flutter compiles it into native apps for Android, iOS, web, and desktop.

The big idea behind Flutter is simple: everything on screen is a widget. Buttons, text, layouts, even the app itself — all widgets, composed together like Lego blocks. You'll see exactly what that means in Lesson 2.

What you need

1. Flutter SDK — download it from flutter.dev and add it to your system PATH. 2. An editor — VS Code with the Flutter extension is the lightest setup and works great. 3. Verify — open a terminal and run `flutter doctor`. It tells you exactly what's missing.

That's it. No heavy IDE required to start learning.

The anatomy of a Flutter app

Every Flutter app follows the same skeleton. Open it up and you'll always find these four pieces:

- `main()` — the entry point. Every Dart program starts here.
- `runApp()` — takes your root widget and draws it on the screen.
- `MaterialApp` — the root widget. It gives your app Material Design styling, navigation, and themes for free.
- `Scaffold` — the basic page structure: an app bar on top, a body in the middle.

Here is the smallest real Flutter app:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'My First App',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('My First App'),
        ),
        body: const Center(
          child: Text('Hello, Flutter!'),
        ),
      ),
    );
  }
}
```

Read it top to bottom: `main()` runs your app, `MaterialApp` sets it up, `Scaffold` builds the page, and inside the body a `Center` widget holds a `Text` widget saying hello. Widgets inside widgets inside widgets — that nesting is the whole game.

Key terms

- `main()` — where every Dart program begins execution.
- `runApp()` — inflates the widget tree and attaches it to the screen.
- `Widget` — the basic building block of a Flutter UI.
- `BuildContext` — a handle to the widget's location in the tree (you'll use it constantly).

Try it yourself

Paste the code above into DartPad (or your editor), press run, and change the text to your own name. Breaking things and fixing them is how this sticks — which is exactly what the exercise below is for.